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An intelligent distributed sensor developed for power cable external damage monitoring

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Abstract

Construction, theft, and geologic changes has long threatened the safety of power cable operation. However, due to the wide distribution of cables and the complicated operating environment, there is currently a lack of effective monitoring for early warning. This paper reports a wireless distributed sensor for monitoring cable external disturbance. Its data processing, low-power power management, energy acquisition unit, and Internet of Things communication methods are described in details; Based on that, the cable vibration test under typical external force excitation are carried out. The time-domain and time-frequency characteristics of the cable vibration signal are obtained and employed as characteristic parameters for external disturbance identification, which are the basis of an intelligent early-warning system for cable antibreakage.

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1. Introduction

In recent years, as the urbanization process has been accelerating, and the core areas of the city have put forward high reliability requirements for power supply, thus the urban power grid cable rate has continued to rise. As a result, the resources of underground cable channels are becoming increasingly tense, and the preventions of fire and external breakage have become essential. The problem of cable damage caused by construction, theft, and

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reconstruction has long threatened the safety of cable operation. However, due to the wide distribution of cables and the complicated operating environment, there is currently no effective monitoring and early warning method. Due to the greater cost of construction and maintenance of traditional online monitoring systems, it has been difficult to apply them in urban cable systems [1].

At present, the distributed optical fiber vibration sensing system based on optical time domain reflectometer technology has been applied in the early warning system, but it is difficult to distinguish the type of disturbance by using the time domain signal, resulting in a high false alarm rate of the system [2-3]. Moreover, the costs of construction and maintenance for full coverage of the cable system is huge. External force disturbances generally cause vibration sound waves to be transmitted on the cable body and the bracket [4]. Therefore, it is a convenient and effective method to monitor the cable vibration sound waves through piezoelectric sensors. It can not only sensitively respond to the vibration intensity, but also perturb according to the vibration mode analysis. However, the SCADA-based monitoring system has high complexity and high maintenance cost, and cannot meet the reliability and economic requirements of the cable feed network [5]. In addition, the amount of cable monitoring data is large, and there are many sensor nodes, which brings great challenges to data storage, information matching, and data applications.

With the proposal and development of Internet of Things (IoT) applications [6-7], a distributed sensor monitoring method based on cable acoustic wave vibration is provided for piezoelectric-based sensing. This paper first develops a wireless distributed sensor based on piezoelectric coupler, and analyzes its data processing, low-power power management, energy acquisition unit, and Internet of Things communication methods; then the cable vibration test is carried out under typical external force excitation. The time-domain and time-frequency characteristics of the cable vibration signal were obtained and employed as the characteristic parameters that could be used for vibration pattern recognition.

2. Design of wireless piezoelectric sensor unit

2.1. Framework of IoT-based low-power sensor

As the basis of the IoT sensor network, the wireless sensor is expected to have a long battery life which is consistent with the power-off maintenance cycle of the monitored object, so the battery life of the sensor node should exceed at last 1 year. In addition, the sensor should also meet the requirements of small size, pluggability and low cost ensuring that it can be deployed on the cable accessories [1]. Though the existing commercial IoT sensors for monitoring temperature, humidity, and acceleration have been adopted in industrial application, they can only convert the physical object value to analog or digital value without advanced data processing. Smart sensing technology specifically for the state of cable external force disturbance needs further exploration; in addition, for cable external force identification, the signal coupling method, vibration characteristics and early warning method also need to be studied in combination with intelligent sensing technology. Therefore, this paper redesigns the traditional online monitoring system based on a MEMS acceleration sensor, and solves the problems of power consumption, sensitivity and timeliness from the aspects of signal coupler, preprocessing unit, MPU, power supply and wireless data transceiver. The hardware architecture of the wireless sensor node designed in this paper is shown in Fig. 1.

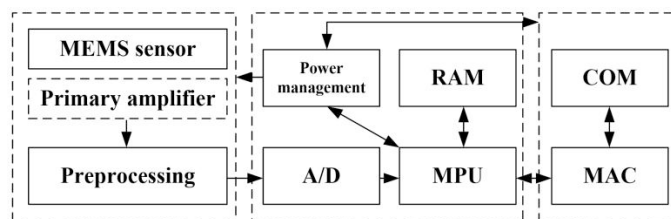


Fig. 1. Hardware framework of wireless sensor node.

It includes three main parts:

Analog signal unit. It is used to expand the sound wave caused by the external force of the cable through a single-axis MEMS acceleration sensor, and to match the dynamic range and analog amplitude of the signal with the A/D sampling unit through active detection amplification processing;

Signal processing and storage unit. It is used to quickly identify effective vibration signals, form minimum feature set (MFS) and sensor status (SS) information, and perform signal processing and data storage. In addition, MPU is also responsible for the processing of downstream command set and power management;

LoRa communication unit. It is used for subcontracting, sending and receiving of upstream and downstream data, and its operation mode is controlled by the signal processing and storage unit. Based on such hardware framework, edge computing and on-site diagnosis can be implemented, reducing the pressure of centralized computing for multipoint data.

2.2. Acoustic coupling module

In the operation of cable, there are several typical types of external disturbance, including air pick signals, mechanical excavation or pile driving signals. The vibration signals caused by external force are mainly distributed in low-frequency range from hundreds hertz to kilohertz. In this paper, a MEMS-based acceleration sensor is selected (ADXL1002) as the acoustic wave coupling frontend, which has a wide linear frequency response (DC-11 kHz@3dB), large measurement range ($\pm 50g$), small temperature coefficient (5%, $-40^{\circ}\text{C}\sim+125^{\circ}\text{C}$), low typical power consumption (1.0 mA), low thermal noise density ($30\text{ }\mu\text{g}/\sqrt{\text{Hz}}$), small size ($<5\text{ mm}^3$) and other advantages, can directly support Internet of Things applications. Figure 2 shows the MEMS acceleration sensor and frequency response characteristic curve.

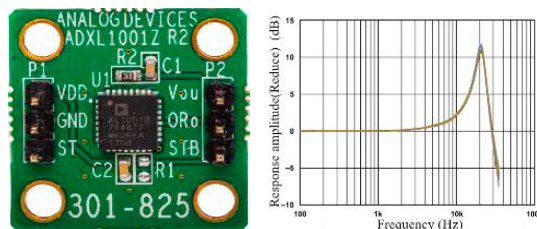


Fig. 2. Acoustic coupling module.

2.3. Signal condition

The main frequency of the vibration signal is in the range of hundreds Hz, and the signal sampling and restoration can be achieved without a very high sampling rate. However, the acceleration amplitude of the vibration signal is usually small. In order to use the storage space reasonably, the vibration signal needs to be amplified by a certain multiple. The amplification circuit is shown in Fig. 3, which is composed of a two-stage amplification circuit composed of LM324. The amplification gain is set to 20dB. In order to eliminate the DC offset, the cascade of amplifiers adopts DC blocking processing. The analog bandwidth of the amplifier unit is 380kHz to ensure that the time-frequency characteristics of the vibration signal are completely retained. Fig. 4 shows the vibration acceleration signal acquired by the piezoelectric sensor unit when an aperiodic signal (cutting the cable) is applied to the cable.

2.4. Sensor power management

This paper uses a current transformer (CT) to draw energy from the cable body to power the sensor energy storage unit. Figure 5 shows a schematic diagram of power capture based on CT. The piezoelectric sensor unit needs a stable constant current source for power supply. Since the sensor needs to work in the monitoring mode for a long

time, the stability and continuity of the power supply is important, and the current carrying capacity of the cable directly affects the power input of the CT coil to draw power. Therefore, in order to avoid the load power supply interruption caused by the reduction of the current carrying capacity (working current is about 150A) or power failure, a lithium battery is used as a bypass backup power supply. In the signal preprocessing unit, a power management circuit is used to stabilize the output level, as shown in Fig. 6. The power management circuit can control the charge and discharge of the battery to prevent the battery from being overcharged and discharged.

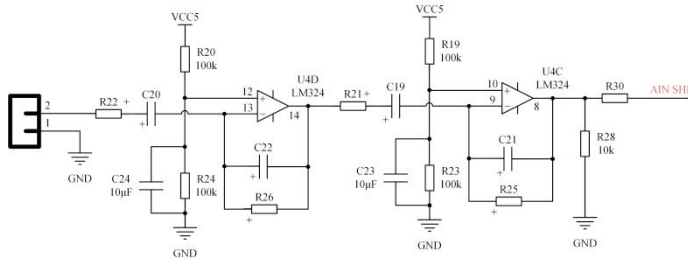


Fig. 3. Amplification circuit of vibration signal.

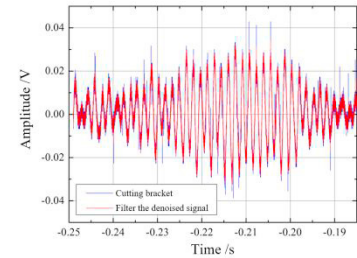


Fig. 4. Aperiodic vibration signal waveform.

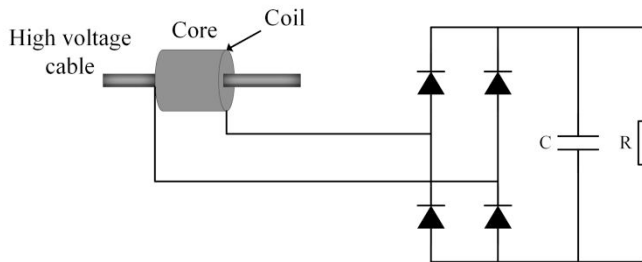


Fig. 5. Power acquiring from high voltage cable.

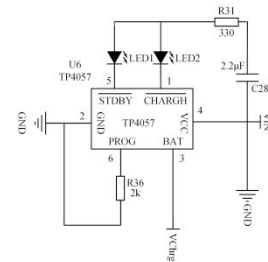


Fig. 6. Power management and voltage level control circuit.

In order to ensure that the output voltage and power meet the needs of the poststage sensors, it is also necessary to design the corresponding power supply voltage regulation and startup circuit, as shown in Fig.7. The control loop adopts the hysteresis comparison method, which improves the control speed and reduces the output voltage ripple. The output voltage of the first-stage pre-regulation circuit is 24V, which is reduced to 5V through the second-stage circuit for the load. The self-starting circuit can take the two-stage output voltage 24V and 5V as input and output the power supply voltage required by the control loop chip. At the same time, the voltage level monitoring is used to start the power output of the lithium battery to the load during the voltage loss start-up, and the self-start circuit is controlled to charge the lithium battery via the Buck regulating circuit.

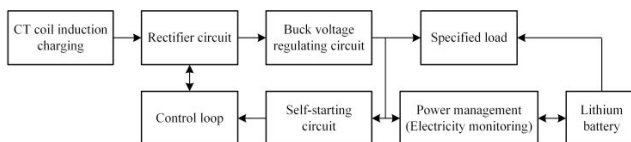


Fig. 7. Operation principle of voltage stabilization and start-up circuit.



Fig. 8. Appearance, internal structure and installation of the sensor.

The appearance and internal structure of the sensor are shown in Fig. 8. The sensor is installed on the base of the shell, which ensures the rigidity of the connection with the shell and the cable. In order to ensure that the shell and

cables of different diameters can be tightly installed, a bakelite material connection piece is used under the shell, and a binding strap is installed on the side.

3. Vibration signal processing and feature extraction

3.1. Uplink/downlink transmission

LoRa modems use proprietary modulation and demodulation procedures to combine spread-spectrum modulation and cyclic error correction coding techniques, thereby increasing link budget and antiinterference. LoRa modulation and demodulation technology can be optimized by adjusting the spreading factor (SF value) and modulation bandwidth (BW). In order to ensure a lower data transmission size and reduce the pressure of the communication unit and centralized calculation, the data transmitted to the gateway through the sensor is the simplified status data. At the same time, in order to meet the requirements of realtime and power consumption, a breakpoint monitoring strategy is applied to practical use. The monitoring density of the sensor N/s is determined by the modulation bandwidth and spreading factor, shown as [10].

$$N = T^{-1} \cdot \frac{SF}{2^{SF} \cdot BW^{-1}} \quad (1)$$

where T is the detection period of the analog signal of the sensor, which is 25 μ s in this article; SF is the spreading factor, 10 is the typical value in this article; BW is the modulation bandwidth, 125 kHz.

3.2. Minimum feature set

Different from the analysis of power frequency voltage signal, the correlation between vibration signal and power frequency excitation is weak, and the vibration excitation is mostly caused by external force disturbance. The randomness of the signal source and the propagation path is strong, and the phase statistical information is difficult to apply. Therefore, this paper uses the quadratic short-time Fourier transform as a kernel function, and performs a time-frequency joint analysis on the vibration transmission signal to find time-frequency characteristic parameters.

If a window function with a short time width is given and the window slides, the quadratic short-time Fourier transform (STFT) $S_z(t, f)$ of the signal $z(t)$ is shown as

$$S_z(t, f) = |STFT_z(t, f)|^2 = \left| \int_{-\infty}^{\infty} z(t') \eta * (t' - t) e^{-j2\pi f t'} dt' \right|^2 \quad (2)$$

$S_z(t, f)$ is the square of the short-time Fourier transform STFT(t, f) mode, which reflects the energy distribution of the signal in the time-frequency plane. Due to the existence of the window function, the quadratic short-time Fourier transform has local characteristics, which are both a function of time and a function of frequency. For a given time t, $S_z(t, f)$ can be regarded as the frequency spectrum at that moment. The quadratic STFT of the discrete time signal $z(n)$ can be shown as

$$S_z(n, \omega) = \left| \sum_{m=-\infty}^{\infty} z(m) \eta(n - m) e^{-i\omega m} \right|^2 \quad (3)$$

The discrete quadratic STFT sampling points at equal frequency intervals can be obtained from

$$S_z(n, r) = \left| \sum_{m=-\infty}^{\infty} z(m) \eta(n-m) e^{-i \frac{2\pi}{N} rm} \right|^2 \quad (4)$$

Let $m = l + n, l = k + sN, k = 0, 1, \dots, N-1$, then equation (7) can be transformed into

$$S_z(n, r) = \left| e^{-i \frac{2\pi}{N} rn} \sum_{l=-\infty}^{\infty} z(l+n) \eta(-l) e^{-i \frac{2\pi}{N} rl} \right|^2 = \left| e^{-i \frac{2\pi}{N} rn} \sum_{k=0}^{N-1} z(k, n) e^{-i \frac{2\pi}{N} rk} \right|^2 \quad (5)$$

Through the above calculation, multiple time-frequency energy centers of vibration non-stationary signals can be obtained, which can be used as one of the characteristic parameters for judging the vibration mode.

In order to ensure a lower data transmission volume, reduce the pressure of the communication unit and the centralized calculation, it is necessary to extract the characteristic value of the cable vibration at sensor side. According to the cable vibration signal, the acceleration, apparent intensity, accumulated energy, event frequency, time-frequency distribution center can be calculated, and the vibration mode and early warning level can be determined. Among them, the apparent intensity and accumulated energy are used as early warning thresholds to start the operation in MPU, and minimum feature set is uploaded to the computer through the communication unit.

In order to use MFS to describe the characteristics of external force disturbance signals, this paper introduces the following state feature quantities based on the time-frequency analysis 1) mean value of vibration cumulative energy (E_{int}); 2) time-frequency energy center (t_m, f_m); 3) 80% energy release duration (t_{80}); 4) Periodic intensity (R_p). The calculation method of the above state characteristic quantities is as follows:

$$E_{\text{int}} = t_1^{-1} \cdot \int_{t_0}^{t_1} \int_0^F S_z(t, f) \quad (6)$$

where t_0 and t_1 are the time when the vibration signal exceeds the measurement threshold for the first time and the monitoring duration, F is the upper limit of the analog bandwidth of the sensor

$$[t_m, f_m]^{(k)} = [S_z^{-1}, \max^{(k)}(S_z(f))] \quad (7)$$

where f_m and t_m are the extreme value of the center frequency of the time-frequency energy density and the corresponding time, and k is the number of extreme values.

$$\int_{t_0}^{t_{80}} \int_0^F S_z(t, f) = 0.8 E_{\text{int}} \quad (8)$$

where t_{80} is 80% energy release duration.

$$R_p = f_{bw} \cdot F^{-1} \quad (9)$$

where f_{bw} is the center frequency bandwidth at 3dB.

Based on the test data, the typical values of the above status characteristic quantities can be obtained, as shown in Table 1. In early warning system, these status quantities can be used as the standards for the identification of cable disturbance patterns and the classification of danger levels, and are continuously accumulated and corrected in practical applications.

Table 1. Characteristic variables of cable vibration at external force disturbances.

Feature name	Typical values			
	electric drill	electric saw	single impulse	tamping
$E_{int}(\text{mV}^2/\text{s})$	10.2~17.3	8.6~11.5	5.5~9.0	4.3~10.4
$(t_m, f_m)^{(3)}$	(1.13, 0.91)	(1.15, 0.86)	(0.97, 1.23)	(3.13, 0.12)
(ms, kHz)	(3.32, 0.90)	(5.12, 0.97)	(2.39, 1.01)	(310.1, 0.12)
	(9.67, 1.05)	(7.38, 0.89)	(2.98, 0.72)	(625.5, 0.13)
$t_{80}(\text{ms})$	/	/	103.4	/
R_p	0.09~0.12	0.07~0.09	0.63~0.82	0.21~0.35

4. Experimental results

Fig.9(a) shows the analysis result of time-frequency characteristics when 100 Hz periodic vibration excitation is applied. It can be seen from the figure that for periodic vibration excitation signals, the vibration amplitude is relatively stable on the time axis, the frequency main component is distributed in the lower frequency band with time, and the high frequency distribution is flat and the amplitude is weak.

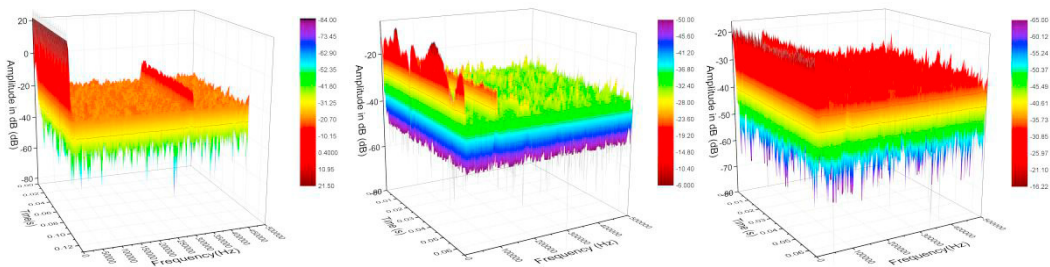


Fig. 9. Time-frequency analysis on signals under 100Hz periodic excitation

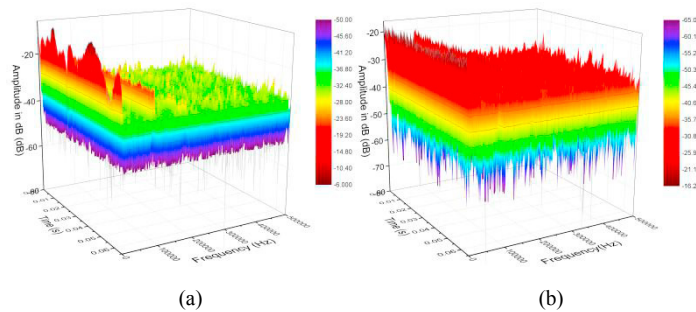


Fig. 10. Time-frequency analysis on signals under electric drill excitations on (b) cable body and (c) cable bracket.

Fig.10(a)(b) shows the analysis result of time-frequency characteristics when single or multiple impact excitation is applied. It can be seen from the figure that for a single tap excitation, the low-frequency principal component of the vibration amplitude exhibits irregular changes on the time axis, the secondary high-frequency component is relatively flat on the time axis, and the higher-frequency secondary component is relatively weak; For continuous ramming excitation, the low-frequency principal component of the vibration amplitude exhibits regular changes on the time axis, and the amplitude at the sub-high frequency is higher than the result of a single tap, and the higher-frequency subcomponent is relatively Strong, the dynamic range of the amplitude is wider.

Fig.11(a)(b) is the result of the time-frequency characteristics of the electric drill excitation transmission. As the vibration of the electric drill is transmitted on the cable body, the low-frequency main component of the vibration

amplitude shows irregular changes on the time axis, the sub-high frequency component is relatively stable on the time axis, and the higher frequency subcomponent is relatively weak; When the vibration of the electric drill is transmitted on the cable bracket, the low-frequency principal component shows regular changes on the time axis, and the characteristics are similar to the former in the sub-high frequency and high-frequency regions. Tab. 2 takes the electric drill applied to the cable as an example, and compares the characteristic quantity obtained by actual measurement data with the process quantity and error output by the sensor. It can be seen from the comparison that the errors between the process quantities used for sensor early warning and the actual measured feature quantities are within 10%, which can meet requirements of the external force monitoring.

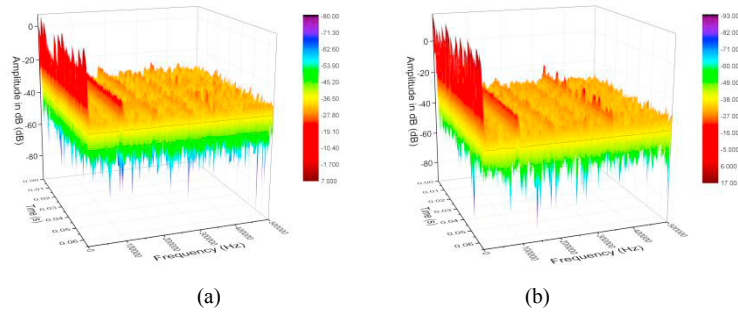


Fig. 11. Time-frequency analysis on signals under electric drill excitations on (d) cable body and (e) cable bracket.

Table 2. Comparisons between actual Procs and sensor output Procs (as electrical drill applied)

Procs	Actual values	Output	Relative error (%)
Proc_acc	13.33	12.90	3.22
Proc_mag	25.51	23.12	9.37
Proc_freq	1	1	0
Proc_Eint	314.25	341.55	8.69
Proc_tmfm	[525.1,13.2]	[520.7,12.6]	[0.8,4.54]
Proc_t80	833	846	1.54
Proc_Rp	0.121	0.133	9.02

5. Conclusions

In this paper a low-power wireless distributed sensor is reported which is developed for cable external force disturbance monitoring. Some key points including data processing, low-power power management, energy acquisition unit, and IoTs communication are discussed in details. By experiment, the time-domain and time-frequency characteristics of the cable vibration signal were obtained and employed as the characteristic parameters that could be extracted at sensor side. Based on that, the minimum feature set are defined for narrow bandwidth communications and concurrent data computing which are the typical features of IoT applications. Finally, it demonstrates that the errors of the process quantities obtained at sensor side are within 10%.

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